

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5112

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 Total Marks:50

Annexure A
CONCEPT MAPPING

Code of Conduct:

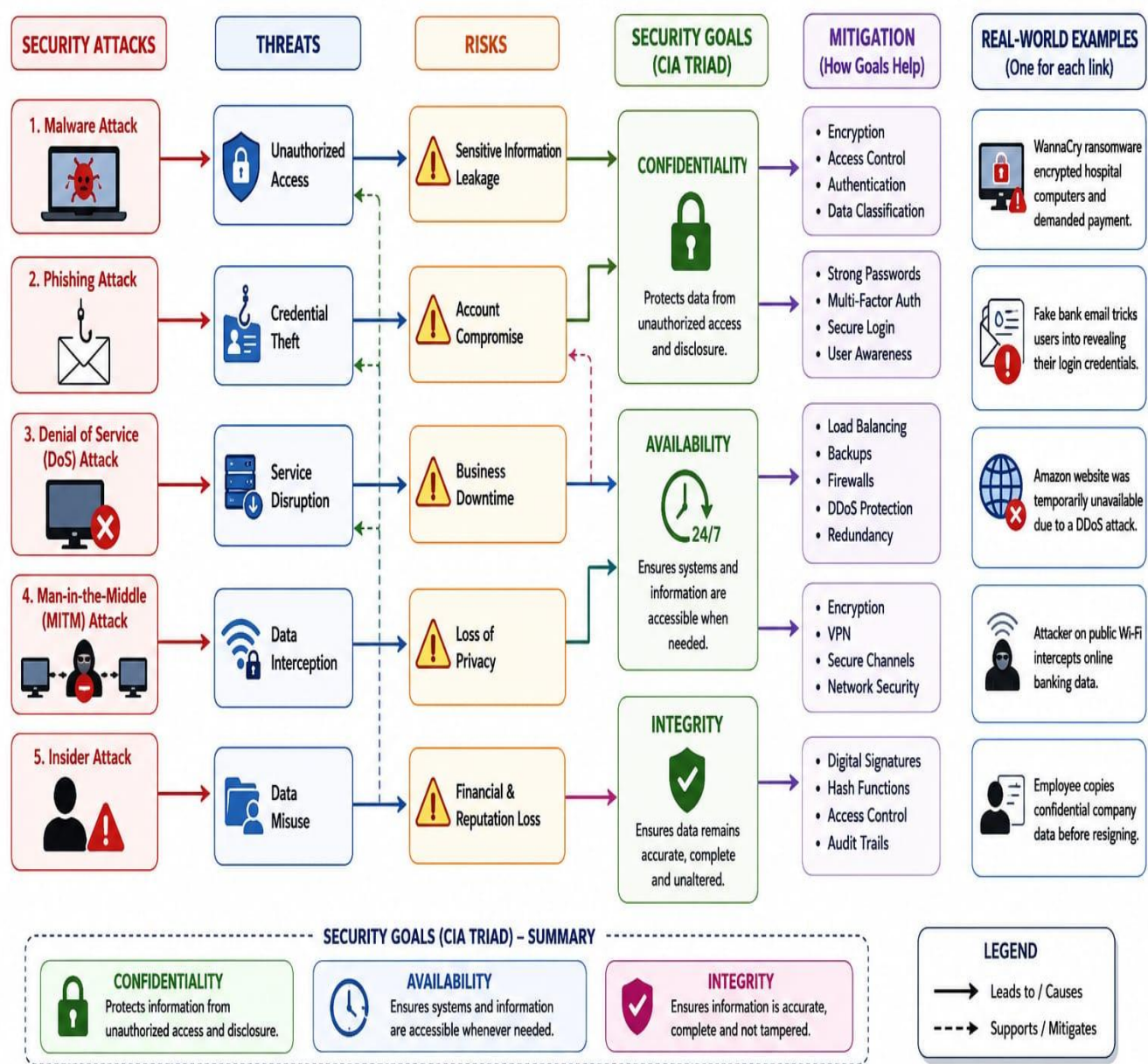
I ABISHEK J(192524131) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
ABISHEK J

Annexure A
CONCEPT MAPPING

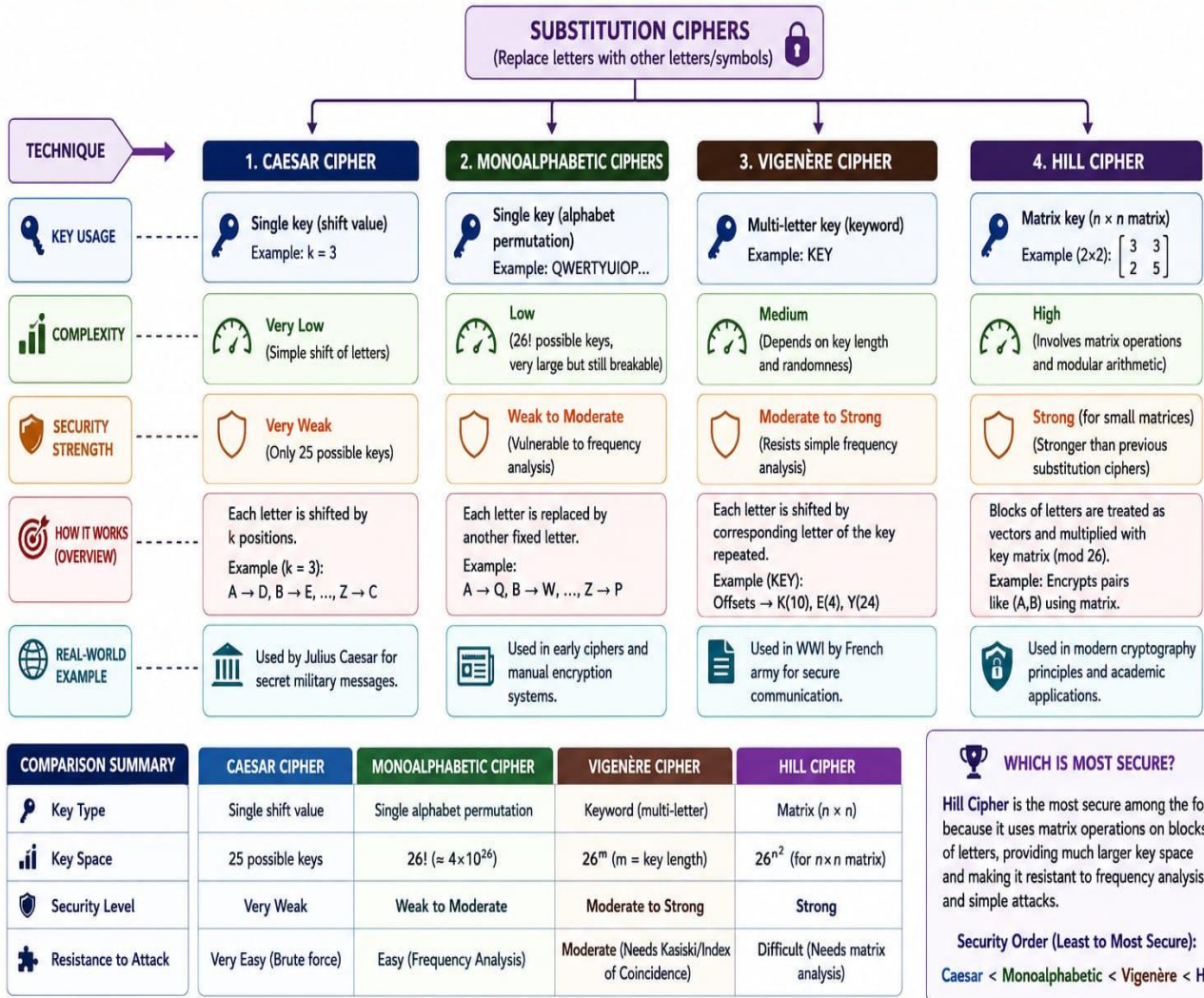
1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
 - a. Show how each attack leads to specific threats and risks
 - b. Map how security goals help mitigate them Provide one real-world example for each link
2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.
3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
 - a. Show differences and similarities
 - i. Map how each affects the Data structure and Security level
 - ii. Analyze which is more vulnerable to attacks and why
4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
 - a. Show relationships between these concepts
 - b. Analyze how each contributes to secure communication
 - c. Include one real-world application for each
5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat’s Little Theorem, Euler’s Phi Function and Euler’s Theorem. Perform the following tasks:
 - a. Show how each concept is mathematically connected
 - b. Map their role in encryption/decryption
 - c. Analyze which concept is most critical for public key cryptography and justify.

CONCEPT MAP – SECURITY ATTACKS, THREATS, RISKS & SECURITY GOALS (CIA TRIAD)



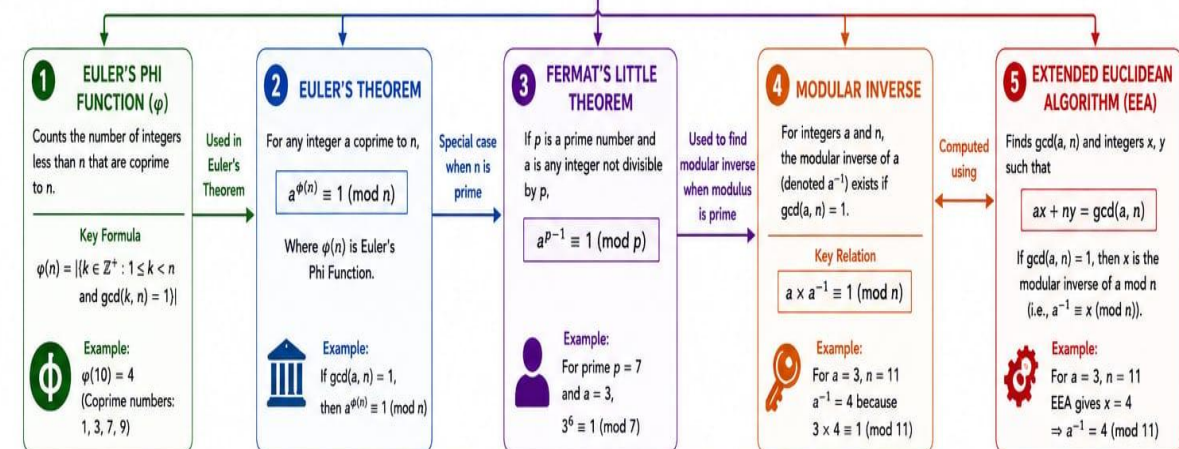
CONCEPT MAP – COMPARISON OF SUBSTITUTION TECHNIQUES

Caesar Cipher • Monoalphabetic Cipher • Vigenère Cipher • Hill Cipher



CONCEPT MAP – NUMBER THEORY CONCEPTS USED IN CRYPTOGRAPHY

Modular Inverse • Extended Euclid Algorithm • Fermat's Little Theorem • Euler's Phi Function • Euler's Theorem



ROLE IN ENCRYPTION / DECRYPTION (RSA EXAMPLE)



MATHEMATICAL CONNECTIONS BETWEEN CONCEPTS

From	Leads To	Mathematical Connection	Explanation
Euler's Phi Function (ϕ)	Euler's Theorem	$a^{\phi(n)} \equiv 1 \pmod{n}$	$\phi(n)$ gives the exponent that makes $a^{\phi(n)} \equiv 1 \pmod{n}$ for $\gcd(a, n) = 1$.
Euler's Theorem	Fermat's Little Theorem	If $n = p$ (prime) $a^{p-1} \equiv 1 \pmod{p}$	FLT is a special case of Euler's Theorem for prime modulus.
Euler's Theorem / Fermat's Little Theorem	Modular Inverse	$a^{\phi(n)-1} \times a \equiv 1 \pmod{n}$ (if $\gcd(a, n) = 1$)	From $a^{\phi(n)} \equiv 1 \Rightarrow$ multiply both sides by a^{-1} to get $a^{-1} \equiv a^{\phi(n)-1} \pmod{n}$. For prime p : $a^{-1} \equiv a^{p-2} \pmod{p}$.
Extended Euclidean Algorithm (EEA)	Modular Inverse	$ax + ny = 1$ $\Rightarrow x \equiv a^{-1} \pmod{n}$	EEA directly gives x such that $ax \equiv 1 \pmod{n}$ when $\gcd(a, n) = 1$.
All Concepts	RSA Security	Large n makes $\phi(n)$ large	Hard to compute d without knowing $\phi(n)$ (prime factorization of n).

WHICH CONCEPT IS MOST CRITICAL FOR PUBLIC KEY CRYPTOGRAPHY?

EXTENDED EUCLIDEAN ALGORITHM (EEA) is the most critical.

- It is the only practical method to compute the modular inverse (d) required in RSA.
- Without EEA, finding d from $e \times d \equiv 1 \pmod{\phi(n)}$ is computationally infeasible for large numbers.
- It also provides $\gcd(a, n)$, which is essential to ensure that inverse exists ($\gcd = 1$).
- Hence, EEA is the backbone of RSA key generation and decryption.

SUMMARY OF ROLES

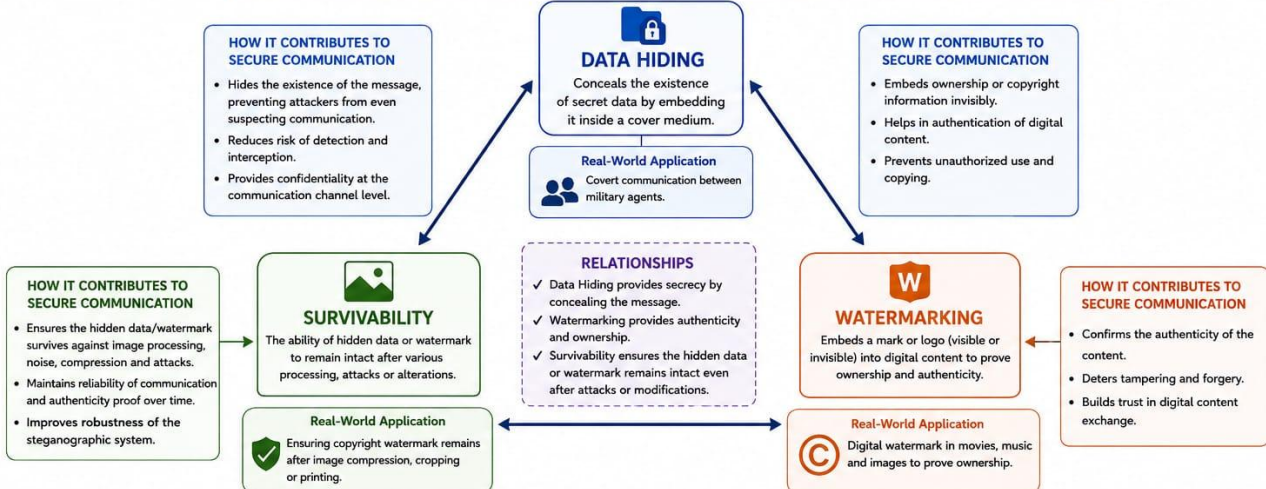
Euler's Phi Function (ϕ)	Counts numbers coprime to n ; used to compute $\phi(n)$ in RSA.
Euler's Theorem	Provides the fundamental relation $a^{\phi(n)} \equiv 1 \pmod{n}$.
Fermat's Little Theorem	Simplifies calculations when modulus is prime.
Modular Inverse	Essential to find private key (d) in RSA.
Extended Euclidean Algorithm	Computes modular inverse efficiently; most critical for RSA.

★ These number theory concepts work together to provide secure and efficient encryption and decryption in modern public key cryptography (e.g., RSA).

🔒 Security relies on the difficulty of factoring large numbers to compute $\phi(n)$ and hence d .

CONCEPT MAP – STEGANOGRAPHY CONCEPTS

DATA HIDING • WATERMARKING • SURVIVABILITY



REAL-WORLD APPLICATIONS (SUMMARY)

DATA HIDING	WATERMARKING	SURVIVABILITY
- Stealth communication in military and intelligence. - Hiding messages inside images for secure transfer. - Covert communication in restricted environments.	- Copyright protection for digital media (images, videos, audio). - Branding and ownership verification. - Tracking illegal distribution of digital content.	- Hidden data remains after compression, resizing, filtering or noise addition. - Essential for real-world usage and reliability. - Ensures long-term authenticity and availability.

OVERALL CONCEPT

Steganography is the art of hiding information in a cover medium. **Data Hiding** ensures secrecy, **Watermarking** ensures ownership and authentication, and **Survivability** ensures the hidden information remains intact even after various attacks or modifications. Together, they enable secure and reliable communication in digital environments.

LEGEND

- Direct Relationship
- - - → Conceptual Relationship

CONCEPT MAP – CLASSICAL ENCRYPTION SCHEMES

SUBSTITUTION TECHNIQUES vs TRANSPOSITION TECHNIQUES

